



UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
Doctoral Program in Deep Tech with a Focus on Artificial Intelligence
and Emerging Technologies

DELIVERABLE No. 1 – PARADIGM JUSTIFICATION STATEMENT

Research topic: Artificial intelligence governance maturity model for Peruvian organizations within the framework of Law No. 31814 and NTP-ISO/IEC 42001:2025: design, validation, and automated compliance assessment.

- 1. Research Topic and Context.** Peru has established an emerging regulatory framework for artificial intelligence, including Law No. 31814, its implementing regulation, and NTP-ISO/IEC 42001:2025. However, there is still no empirical evidence showing how prepared Peruvian public and private organizations are to govern AI under this framework, nor a maturity model validated in the Peruvian context. This gap matters because organizations need a measurable way to identify their current governance maturity, compliance gaps, and priorities for improvement. The population of interest is Peruvian public and private organizations that deploy, manage, or govern AI systems.
- 2. Preliminary Research Question.** How can an AI governance maturity model, contextualized to the Peruvian regulatory framework, be designed and validated to enable public and private organizations to assess and improve their effective compliance capacity, while also supporting a reliable automated assessment of documentary evidence through large language models?
- 3. Chosen Paradigm and Justification.** This research adopts Design Science Research as its primary paradigm, supported by an exploratory sequential mixed-methods design. Design Science is the best fit because the doctoral contribution is not limited to observing or explaining AI governance maturity; it explicitly constructs and evaluates artifacts: a maturity model, a validated assessment instrument, and an LLM-based automated compliance assessment artifact. I reject a purely positivist paradigm because statistical measurement of maturity levels and associated factors, although essential for validation, would not by itself produce the designed governance solution required by the research problem. I reject a purely interpretivist paradigm because understanding organizational meanings and expert perspectives alone would not provide the psychometric validation, national baseline, or measurable agreement required by the study. I also reject mixed methods as the primary paradigm because combining qualitative and quantitative evidence describes how the evidence will be integrated, but the central doctoral contribution remains the purposeful construction and rigorous evaluation of artifacts. The exploratory sequence supports Design Science: literature and expert knowledge inform the model, quantitative validation and organizational application test it, and the LLM artifact is subsequently evaluated against human experts.
- 4. Implications of My Choice.** This choice means that (a) my data will include systematic-review evidence, expert judgments, maturity-instrument responses from approximately 60–80 Peruvian organizations, organizational governance documents, and human-versus-LLM assessment results; (b) my methods will include PRISMA 2020, Delphi, Aiken's V, Cronbach's alpha, organizational survey analysis, artifact development, and Cohen's kappa; (c) my contribution will be a validated Peruvian AI-governance maturity model, a national empirical baseline, and an automated assessment artifact; and (d) appropriate publication targets include Information Systems Management and Internet Research, both aligned with the governance and information-systems domain represented in the project's literature base.
- 5. One Doubt or Tension.** My main methodological tension is access to senior organizational informants and sufficiently rich governance documentation, because limited or uneven evidence could weaken both the national baseline and the evaluation of the LLM artifact. I will therefore treat access and documentary completeness as an explicit study limitation and use the planned multi-organization sample, expert validation, and human–LLM comparison to strengthen the credibility of the findings.

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